Conclusions

1. Elliptical Curve vs Finite-Field

In comparing Elliptic Curve Cryptography to the traditional finite-field Diffie-Hellman key exchange (DHKE), the core mathematical mechanisms differ significantly. Finite-field Diffie-Hellman relies on modular exponentiation to hide the private exponent, requiring an attacker to solve the standard Discrete Logarithm Problem. In contrast, Elliptic Curve Diffie-Hellman (ECDH) uses scalar multiplication, adding points on an elliptic curve multiple times, which relies on the Elliptic Curve Discrete Logarithm Problem (Raza, 2026). Because there are no known sub-exponential algorithms to solve this like there are for finite fields, ECDH provides the exact same security with a much smaller key size (Rabah, 2006).

2. Why use 16-32 bit are insecure

The sizes are used for simulation and is not secure due to the small-size pool that it has, which is 65,536. This is insecure because computers can brute force this in seconds, so it would need at 10x the bit size to even think of resistance of brute force.

3. Factoring p-1 limits implantation

We used a basic trial division to find the factors to test against the distinct prime of p-1. The project uses trail division to find the factors used in the simulation. Factoring the large integers is difficult, so that is why for a simulation-based project. It is more preferred to use a simple method to understand the process of the concept that is DHKE.

4. What is Ephemeral ECDH

It means temporary. In ECDHE, Alice generates a new and unique key pair forever single session rather than reusing an old key to compute shared secret. Once the session ended, the private keys are deleted and discarded after its use.

5. Fresh session keys and how they relate to forward secrecy

If its forward secrecy means that an attacker, even with a stolen long-term key, still cannot decrypt past sessions. Because the actual session uses ephemeral keys that were thrown after use. So even with identity key stolen the past logs are safe.

6. Why ECDH still need authentication
   Because elliptical curve and finite-field does not have any idnetity verification, it only provides key exchange. So without them, in the simulation, Eve was acting as a man-

in-the-middle attack. Showcasing that Eve can intercept and gets information from Bob and Alice wihtouth the participants knowing. Showcasing the ED25519 keys or the concept of authentication and signing are vital in exchanging messages online.

7. Why production apps used established library's and protocols.

   Implementing these functions from scratch is highly vulnerable that could be missed due to the lack of multiple reviewers. Allowing force attacks with timing or brute force methods. The established protocols and libraries have been tested and standardized by experts in the field to allow them to be used safely and secured in their algorithm sound logic.

Reference

Kefa . Rabah, 2006. Elliptic Curve Cryptography over Binary Finite Field GF(2^m). *Information Technology Journal, 5: 204-229.*

Raza, Muhammad. “Elliptic Curve Cryptography: An Introduction.” *Splunk*, www.splunk.com/en_us/blog/learn/elliptic-curve-cryptography.html. Accessed 30 Oct. 2026.